

COMP244 — Emotion Classification Project Report

1. Introduction

Emotion classification is a natural language processing (NLP) task that involves assigning emotional categories to text samples. This project investigates the performance of several classical machine learning models on multi-class emotion classification using TF-IDF feature extraction.

The system was developed using traditional NLP preprocessing techniques combined with supervised learning algorithms. The primary objective was to determine which model generalises best under varying amounts of training data while maintaining strong performance on minority emotion classes.

2. Dataset Description

The dataset contains 8,000 labelled text samples distributed across seven emotion categories: sadness, neutral, worry, happiness, fun, surprise, and hate.

Exploratory analysis confirmed that the dataset contains no missing values or duplicate entries. However, the class distribution is imbalanced, particularly for the surprise category. Because of this imbalance, macro F1 score was treated as a primary evaluation metric alongside accuracy and weighted F1.

3. Preprocessing Pipeline

The preprocessing pipeline applies several transformations to normalize raw text before feature extraction:

- Lowercasing
- URL removal
- Punctuation and number removal
- Whitespace normalization

These preprocessing steps reduce noise and improve vocabulary consistency, which is especially important for sparse TF-IDF representations.

4. Feature Extraction

Text samples were transformed into numerical vectors using TF-IDF (Term Frequency–Inverse Document Frequency).

The vectorizer was configured with:

- stop_words='english'
- ngram_range=(1,2)
- max_features=10000
- sublinear_tf=True
- min_df=2

These settings improve generalisation by removing noisy tokens, capturing short phrases through bigrams, and limiting extremely rare features. The TF-IDF vectorizer was always fitted exclusively on training data to avoid data

leakage.

5. Machine Learning Models

Four classical machine learning algorithms were evaluated:

Decision Tree:

Decision trees recursively partition the feature space using information gain. Although interpretable, they perform poorly on high-dimensional sparse TF-IDF representations and tend to overfit.

LinearSVC:

LinearSVC searches for a hyperplane that maximizes class separation margins in the feature space. The final model uses LinearSVC, which is a linear Support Vector Machine. It does not apply the kernel trick. This design choice is appropriate because TF-IDF generates extremely high-dimensional sparse feature vectors in which linear classifiers perform effectively while remaining computationally efficient.

Compared with kernel-based SVMs, LinearSVC scales significantly better to large sparse vocabularies and requires substantially less computational resources.

k-Nearest Neighbours:

kNN classifies samples using nearest neighbours according to cosine similarity. While cosine distance is suitable for sparse text vectors, kNN still suffers from the curse of dimensionality.

Multinomial Naive Bayes:

Naive Bayes applies probabilistic classification under a conditional independence assumption. Despite this assumption being unrealistic for natural language, the model remains highly competitive for many NLP tasks.

Ensemble Methods:

Voting-based ensembles were explored during experimentation to combine predictions from multiple classifiers. However, ensembles did not consistently outperform the standalone LinearSVC model and introduced additional complexity without significant macro F1 improvements.

6. Experimental Setup

Three stratified train/test splits were evaluated:

- 40% train / 60% test
- 30% train / 70% test
- 20% train / 80% test

Stratified splitting preserved class distributions across all experiments. The TF-IDF vectorizer was refitted only on the training partition for each split.

7. Evaluation Metrics

The following metrics were computed:

Accuracy:

Measures overall classification correctness.

Macro F1:

Computes the unweighted average F1 score across all classes and is especially important for imbalanced datasets.

Weighted F1:

Computes a class-frequency-weighted average of F1 scores.

8. Results and Discussion

LinearSVC achieved the highest performance across all split configurations. This result is consistent with the strong performance of linear classifiers in high-dimensional sparse text spaces.

Naive Bayes provided competitive baseline performance despite its simplifying assumptions. kNN achieved moderate performance but remained limited by sparse high-dimensional geometry. Decision Tree performed worst due to overfitting tendencies in TF-IDF feature spaces.

As training data decreased, all models exhibited reduced macro F1 performance, especially on minority classes such as surprise.

9. Final Model

The final selected model is a pipeline consisting of TF-IDF vectorization followed by LinearSVC.

This model was selected because it consistently achieved the highest macro F1 scores while remaining computationally efficient and robust to class imbalance. Balanced class weights were used to improve minority class performance.

Although ensemble methods were explored, they did not provide sufficient improvements to justify the added complexity. Therefore, the final implementation retains the simpler LinearSVC-based architecture trained on the full dataset.

10. Conclusion

This project demonstrated that TF-IDF combined with LinearSVC is highly effective for multi-class emotion classification using classical machine learning techniques.

The experiments confirmed that:

- linear classifiers perform exceptionally well on sparse TF-IDF text data,
- macro F1 is essential for imbalanced datasets,
- simpler models can outperform more complex alternatives,
- ensemble methods do not necessarily guarantee improved performance.

Future work may include transformer-based models such as BERT and advanced contextual embeddings for improved semantic understanding.